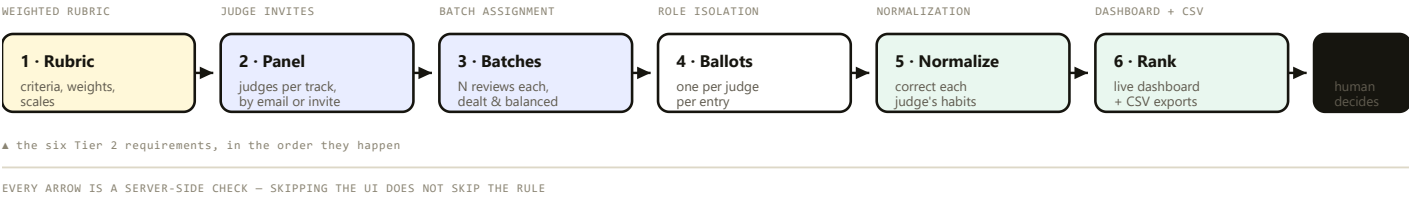


Tier 2 is the part that decides who wins. It takes a pile of submitted entries and a panel of humans who disagree, and turns them into a ranking an organizer can defend. Six requirements, all shipped, all enforced by the API — not by the web page.



1 The rubric: what a ballot actually is

The organizer defines the criteria. Each has a **weight** (relative, not a percentage — 3/1/1 ranks exactly like 60/20/20) and its own **scale**. A ballot must score *every* criterion, inside its own range, and nothing the rubric does not define. The server rejects anything else.

A BALLOT ON ONE ENTRY

SCORE ÷ MAX × WEIGHT

Technical depth
weight 3 · out of 10

9

9/10 × 3 = 2.70

Originality
weight 2 · out of 10

7

7/10 × 2 = 1.40

Completeness
weight 2 · out of 10

8

8/10 × 2 = 1.60

Weighted score

(2.70 + 1.40 + 1.60) ÷ 7

81.4%

Total weight 3+2+2 = 7. Dividing by it lands every ballot in 0–100%.

weighted = Σ (score ÷ max × weight) ÷ Σ weight

Why divide by each criterion's own max first? Without it, a 1–10 criterion would silently count twice as much as a 1–5 criterion carrying the same weight. Scaling first means the weight is the *only* thing that decides influence.

An entry's raw score is the plain mean of its ballots' weighted scores. That is the number before any correction for who did the judging — which is what **Stage 5** on page 3 exists to fix.

missing a criterion → 400

score above max → 400

unknown criterion → 400

VERDIKT • TIER 2 REPORT

FULL METHOD: JUDGING.MD • 210 TESTS • STAGEF–STAGEH

2 Getting judges onto the panel

Judges are appointed **per track**, two ways: by the email on an existing account, or by an invite link for someone who has not signed up yet. A judge link is a credential, so it is stricter than a team link.



BOUND TO ONE ADDRESS — A FORWARDED LINK IS USELESS • SINGLE USE • REVOCABLE • EXPIRES

Nobody who is competing in or organizing the event can be appointed — and the check runs *again* the moment an invite is accepted, so someone who joined a team after being invited is still refused. Admin status grants nothing: admins are staff, not panel members.

3 Scope: a judge scores only what they were given

This is the rule that makes "a track judge cannot open another track" true. Scope narrows in three steps — the server picks the first that applies:

1 Batches exist

Scores exactly the entries assigned to them.
Nothing else — not even in their own track.

2 No batches yet

Scores entries in the tracks they were appointed to. Other tracks are refused.

3 On the event, no track

Scores every entry. The fallback for a small event with no track split.

WHAT EACH ROLE CAN READ

Judge

own ballots

peer ballots X

standings X

Organizer

their own event only

another organizer's event X

WHY A JUDGE CANNOT SEE STANDINGS: THE AGGREGATE PLUS YOUR OWN BALLOTS SOLVES FOR EVERYONE ELSE'S.

4 Batch assignment: "3 independent reviews per entry"

The organizer picks a number; the server deals it. Four rules, in priority order:

TRACK: SECURITY

Entry A → Ada, Bo, Cy

Entry B → Ada, Bo, Cy

TRACK: INFRA — ONLY 2 JUDGES

Entry C → Dee, Eli

Entry D → Dee, Eli

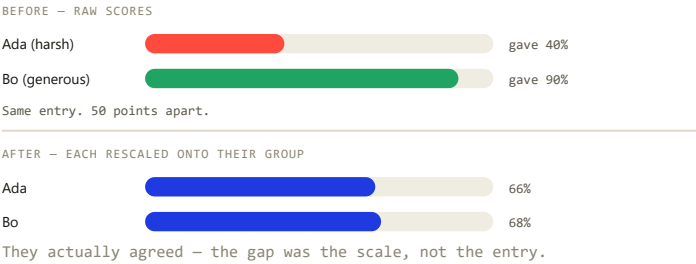
Shortfall reported: 2 of 3
It will NOT borrow Ada from Security.

RULE	WHAT IT MEANS
1 Respect tracks	An entry only goes to judges of its own track.
2 Adopt ballots	Already scored it? It joins their batch — nobody is locked out of a ballot they can stand behind.
3 Balance load	Hardest entries dealt first, each to the least-loaded eligible judges. Deterministic.
4 Report shortfall	Too few judges in a track? Say so. Never quietly fill the gap from elsewhere.

Removal narrows, never widens. Taking a judge off their last track also removes them from the event and drops their unscored assignments — and re-dealing will not hand the track back. (A real bug we found and fixed: removal used to make them an *event-wide* judge.)

5 Cross-judge normalization

The problem: judges disagree about the scale, not just the entry. With three reviews each, which judges an entry happened to draw can matter more than the entry itself.



$$\text{normalized} = \text{groupMean} + \left(\frac{\text{weighted} - \text{judgeMean}}{\text{judgeSd} \times \text{groupSd}} \right)$$

For each judge, the server takes their **mean** and **spread** across every ballot they cast, strips both out, then puts the result back on the group's scale — so it still reads as a percentage. Removing the mean fixes *harsh vs generous*; removing the spread fixes *decisive vs everything-is-a-7*.

Why "group", not the whole panel? Bias is only measurable between judges who saw *overlapping work*. Verdict links judges who share an entry — directly or through a chain — and rescales within each group. Rescaling onto the whole panel would drag a genuinely stronger track down to the average.

LEFT UNCORRECTED WHEN...	WHAT HAPPENS INSTEAD	WHY
the judge cast fewer than 2 ballots	ballot stands as cast	one ballot can't tell a harsh judge from a weak entry
the judge shares no entry with anyone	ballot stands as cast	there is nobody to compare them against
every ballot they cast is identical	set to the group mean	they expressed no preference between entries

Does it actually work? Measured, not claimed.

`npm run normalization-proof` simulates 200 events per scenario where the *true* ranking is known, and scores each method against it (1.000 = perfect).

SCENARIO (40 ENTRIES, 8 JUDGES, 3 REVIEWS)	RAW	TEXTBOOK Z	VERDIKT	VERDICT
Biased judges, entries dealt at random	0.935	0.975	0.975	clearly helps — beats raw in 97% of events
Fair judges, entries dealt at random	0.986	0.977	0.977	costs a little; raw is one click away
Biased judges kept to tracks of differing strength	0.903	0.796	0.906	textbook method breaks; grouping is why ours doesn't

Row 3 is the row that decided the design — and the honest one: it is why Verdict does not ship the textbook z-score.

6 Publishing: the dashboard and the exports

● **Live dashboard**

Rankings overall *and* per track, refreshed every few seconds. Raw and normalized shown side by side — the organizer picks which one ranks, the other stays visible. Computed from ballots on every read, never stored, so it is never stale.

● **Honest ranks**

Ties share a rank (1, 2, 2, 4) — float noise within 10^{-9} cannot invent a winner. **An entry nobody scored is unranked**, listed last with no score — unjudged is not the same as last. Drafts never rank.

● **CSV at every stage**

Entries (drafts + answers) · assignments (who owes what) · every ballot (raw + normalized) · final standings. Organizer-only. Cells are quoted and defused, so `=HYPERLINK(...)` opens as *text*, not a formula.

What Verdict deliberately does not do: it ranks, it does not award. There is no "publish results" step — the last call stays with a human, and the report above is what they use to defend it.